

UNIVERSITY LAB REPORT

Department of Electrical Engineering (Communications)

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Reg. No:	0672	Subject:	Programming Fundamentals (C++)
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Objectives

The objective of Lab 13 is to understand and implement C++ structures (structs). A structure is a user-defined data type that groups related variables of different types under one name. In Task 1, a Person structure is declared and initialized. In Task 2, a Student structure is used to store academic information. In Task 3, a Book structure demonstrates accessing members of multiple structure variables using dot notation.

Task 1: Structure Declaration and Initialization — Person

Description: A structure Person is declared with three members: name (string), age (integer), and address (string). A variable person1 of type Person is created, its members are initialized with values, and then displayed using the dot (.) notation.

■ C++ Source Code

```
// Input/output aur string ke liye zaruri header files include kar rahe hain
#include <iostream>
#include <string>

// std namespace use kar rahe hain
using namespace std;

// 'Person' naam ki structure define kar rahe hain
struct Person {
    string name; // Member 1: insaan ka naam store karega
    int age; // Member 2: insaan ki umar store karega
    string address; // Member 3: insaan ka pata store karega
};

int main() {
    // Person type ka variable 'person1' declare kar rahe hain
    Person person1;

    // Dot notation se person1 ke har member ko value assign kar rahe hain
    person1.name = "Shah Khalid";
```

```

person1.age = 21;
person1.address = "House No. 5, Street 3, Risalpur Cantt, KPK";

// person1 ke members ki values screen par print kar rahe hain
cout << "==== Person Information =====>> endl;
cout << "Name : " << person1.name << endl;
cout << "Age : " << person1.age << " years" << endl;
cout << "Address : " << person1.address << endl;

return 0; // Program successfully khatam
}

```

■ Program Output

```

==== Person Information =====
Name : Shah Khali
Age : 21 years
Address : House No. 5, Street 3, Risalpur Cantt, KPK

```

Explanation: The struct keyword creates a user-defined data type. Each structure variable (like person1) gets its own copy of all the members. The dot operator . is used both to assign values to members and to read them.

Task 2: Structure Declaration and Initialization — Student

Description: A structure Student is declared with three members: name (string), age (integer), and grade (char). A variable student1 is created, initialized, and all member values are displayed on the screen.

■ C++ Source Code

```

// Zaruri header files include kar rahe hain
#include <iostream>
#include <string>

using namespace std;

// 'Student' structure define kar rahe hain teen members ke saath
struct Student {
    string name; // Student ka naam
    int age; // Student ki umar
    char grade; // Student ka grade (single character: A, B, C ...)
};

int main() {

    // Student type ka variable 'student1' declare kar rahe hain
    Student student1;

    // Dot notation se student1 ke har member ko value de rahe hain
    student1.name = "Shah Khalid";
    student1.age = 21;
    student1.grade = 'A'; // Single quotes char ke liye use hote hain
}

```

```
// student1 ke saare members screen par display kar rahe hain
cout << "==== Student Information =====" << endl;
cout << "Name : " << student1.name << endl;
cout << "Age : " << student1.age << " years" << endl;
cout << "Grade : " << student1.grade << endl;

return 0;
}
```

■ Program Output

```
==== Student Information =====
Name : Shah Khali
Age : 21 years
Grade : A
```

Explanation: The char data type stores a single character and uses single quotes. Notice how the same dot notation is used for all member types — string, int, and char — making structures flexible containers for mixed data.

Task 3: Accessing Structure Elements — Book (book1 & book2)

Description: A structure Book is declared with four members: title (string), author (string), price (float), and pages (integer). Two variables, book1 and book2, are declared, initialized with different values, and both are displayed to show how one structure type can hold multiple independent records.

■ C++ Source Code

```
// Zaruri header files include kar rahe hain
#include <iostream>
#include <string>

using namespace std;

// 'Book' structure define kar rahe hain char members ke saath
struct Book {
    string title; // Kitaab ka naam
    string author; // Kitaab likhne wale ka naam
    float price; // Kitaab ki qeemat (decimal number ho sakti hai)
    int pages; // Kitaab mein total safhe
};

int main() {

    // Pehli kitaab ke liye 'book1' variable declare kar rahe hain
    Book book1;

    // book1 ke members initialize kar rahe hain dot notation se
    book1.title = "The C++ Programming Language";
    book1.author = "Bjarne Stroustrup";
    book1.price = 1850.50f; // 'f' batata hai ke yeh float value hai
}
```

```

book1.pages = 1376;

// Doosri kitaab ke liye 'book2' variable declare kar rahe hain
Book book2;

// book2 ke members initialize kar rahe hain
book2.title = "Let Us C";
book2.author = "Yashavant Kanetkar";
book2.price = 650.00f;
book2.pages = 675;

// book1 ki tamam details screen par print kar rahe hain
cout << "===== Book 1 Details =====" << endl;
cout << "Title : " << book1.title << endl;
cout << "Author : " << book1.author << endl;
cout << "Price : Rs. " << book1.price << endl;
cout << "Pages : " << book1.pages << endl;

// book2 ki tamam details screen par print kar rahe hain
cout << "\n===== Book 2 Details =====" << endl;
cout << "Title : " << book2.title << endl;
cout << "Author : " << book2.author << endl;
cout << "Price : Rs. " << book2.price << endl;
cout << "Pages : " << book2.pages << endl;

return 0; // Program successfully mukammal ho gaya
}

```

■ Program Output

```

===== Book 1 Details =====
Title : The C++ Programming Language
Author : Bjarne Stroustrup
Price : Rs. 1850.5
Pages : 1376

===== Book 2 Details =====
Title : Let Us C
Author : Yashavant Kanetkar
Price : Rs. 650
Pages : 675

```

Explanation: A single structure definition (Book) can be used to create as many independent variables as needed. Each variable — book1 and book2 — holds its own separate data. The float suffix f ensures the value is stored as a single-precision floating-point number.

Key Concepts Summary

Concept	Syntax Example	Purpose
Structure Declaration	<code>struct Person { ... };</code>	User-defined composite data type banata hai

Variable Declaration	<code>Person person1;</code>	Structure type ka variable create karta hai
Member Assignment	<code>person1.name = "Ali";</code>	Dot notation se member ko value deta hai
Member Access	<code>cout << person1.age;</code>	Dot notation se member ki value padhta hai
char Member	<code>char grade = 'A';</code>	Single character store karta hai
float Member	<code>float price = 650.0f;</code>	Decimal number store karta hai
Multiple Variables	<code>Book book1; Book book2;</code>	Ek structure se kai alag records bana sakte hain

Conclusion

In Lab 13, the concept of C++ structures was studied and implemented through three practical tasks. Task 1 demonstrated the Person structure with string and integer members, showing basic declaration, initialization, and display using dot notation. Task 2 extended this with the Student structure, introducing the char data type for the grade member. Task 3 used the Book structure with a float member and two separate variables (book1, book2), demonstrating how a single structure definition can create multiple independent records. All programs were written with line-by-line Urdu comments in English letters. Structures are the foundation of object-oriented thinking and are widely used in real-world C++ applications.

Student Signature: _____

Instructor Signature: _____